TauTimp — Spherical Symmetry: Generalized Variable-Coefficient Wave Equation

Author: Viorel Popescu **Framework:** Teoria Ondusculară (TauTimp) **Parameter:** $R = 4.2355 \times 10^{22}$ m

1. Master Equation

The TauTimp spherical-symmetry field equation is:

$$\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,t)}{\partial t^2}$$

In full spherical coordinates the Laplacian is:

$$\Delta = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

With **azimuthal symmetry** ($\partial/\partial\phi = 0$) we write $H(r,\theta,t) = f(r,t)\,Y_\ell^0(\theta)$ where $Y_\ell^0(\theta) = P_\ell(\cos\theta)$ is the Legendre polynomial of degree ℓ.

2. Reduced Radial–Time Equation

Substituting the ansatz and using $\Delta P_\ell = -\ell(\ell+1)P_\ell/r^2$ for the angular part, the scalar field $f(r,t)$ satisfies the **variable-coefficient wave equation**:

$$\boxed{\frac{R}{r}\left[\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} - \frac{\ell(\ell+1)}{r^2}f\right] = \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}}$$

For the **spherically symmetric mode** ($\ell = 0$):

$$\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}\right) = \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}$$

Defining $\tilde{f}(r,t) = r\,f(r,t)$ converts the radial part to:

$$\frac{R}{r}\frac{\partial^2 \tilde{f}}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 \tilde{f}}{\partial t^2}$$

3. Separation of Variables

Let $f(r,t) = g(r)\,u(t)$. Substituting:

$$\frac{R}{r}\left(g''u + \frac{2}{r}g'u\right) = \frac{g\,u''}{V^2}$$

Dividing by $g(r)u(t)$:

$$\frac{R}{r}\cdot\frac{g'' + \frac{2}{r}g'}{g} = \frac{1}{V^2}\cdot\frac{u''}{u} = -k^2 \quad \text{(separation constant)}$$

This gives two ODEs:

3.1 Temporal ODE

$$u'' + k^2V^2\,u = 0 \Longrightarrow u(t) = A\cos(\omega t) + B\sin(\omega t), \quad \omega = kV$$

3.2 Radial ODE

$$g'' + \frac{2}{r}g' + \frac{k^2 r}{R}\,g = 0$$

Substituting $g(r) = r^{-1/2}\,w(r)$:

$$w'' + \frac{1}{r}w' + \left(\frac{k^2 r}{R} - \frac{1}{4r^2}\right)w = 0$$

This is a **Bessel-type equation**. With the further substitution $\xi = \frac{2k}{3}\sqrt{\frac{r^3}{R}}$ (valid for large r), or equivalently letting $\zeta = \left(\frac{3k}{2\sqrt{R}}\right)^{2/3} r$, the equation reduces exactly to the **Airy equation**:

$$\frac{d^2G}{d\zeta^2} - \zeta\,G = 0$$

where $G(\zeta) = r^{1/2}g(r)$.

3.3 Airy Function Solution

The general radial solution is:

$$g(r) = r^{-1/2}[C_1\,\mathrm{Ai}(-\zeta(r)) + C_2\,\mathrm{Bi}(-\zeta(r))]$$

with the dimensionless variable:

$$\zeta(r) = \left(\frac{3k}{2}\right)^{2/3}\left(\frac{r}{R^{1/3}}\right), \qquad \text{i.e.} \quad \zeta(r) = \left(\frac{3k}{2\sqrt{R}}\right)^{2/3} r$$

4. Boundary Conditions and Eigenvalue Quantization

Case A: $f(2R,0) = 0$

The condition $g(2R) = 0$ requires:

$$C_1\,\mathrm{Ai}(-\zeta(2R)) + C_2\,\mathrm{Bi}(-\zeta(2R)) = 0$$

Setting $C_2 = 0$ (physically motivated — Bi diverges), the eigenvalues satisfy:

$$\mathrm{Ai}(-\zeta_n(2R)) = 0 \Longrightarrow \zeta_n(2R) = a_n \quad (n = 1,2,3,\dots)$$

where a_n are the n-th positive zeros of $\mathrm{Ai}(-x)$, i.e. $a_1 \approx 2.3381,\ a_2 \approx 4.0879, \dots$

The **quantized wave numbers** are:

$$k_n = \frac{2}{3}\left(\frac{a_n\sqrt{R}}{(2R)}\right)^{3/2} \cdot \frac{1}{\sqrt{R}} = \frac{2}{3\,(2R)}\left(\frac{a_n}{1}\right)^{3/2}$$

More explicitly:

$$k_n = \frac{2}{3}\,\frac{a_n^{3/2}}{(2R)^{3/2}} \cdot R^{1/2} = \frac{2\,a_n^{3/2}}{3\,(2R)^{3/2}}\,R^{1/2}$$

$$\boxed{k_n = \frac{a_n^{3/2}}{3 \cdot 2^{1/2}\,R}, \qquad \omega_n = k_n V}$$

Case B: $f(R,0) = 0$

By the same argument with boundary at $r = R$:

$$k_n^{(R)} = \frac{2\,a_n^{3/2}}{3\,R^{3/2}} \cdot R^{1/2} = \frac{2\,a_n^{3/2}}{3\,R}$$

$$\boxed{k_n^{(R)} = \frac{2\,a_n^{3/2}}{3\,R}, \qquad \omega_n^{(R)} = k_n^{(R)} V}$$

5. Full Field Solution $f(r,t)$

Superposing all eigenmodes (Case A, $\ell = 0$):

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} \frac{1}{\sqrt{r}}\,\mathrm{Ai}(-\zeta_n(r))[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)]}$$

Coefficients A_n, B_n are fixed by initial conditions $f(r,0)$ and $\partial_t f(r,0)$.

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6. Full Angular Field $H(r,\theta,t)$

Expanding in Legendre polynomials:

$$\boxed{H(r,\theta,t) = \sum_{\ell=0}^{\infty}\sum_{n=1}^{\infty} \frac{D_{\ell n}}{\sqrt{r}}\,\mathrm{Ai}\left(-\zeta_n^{(\ell)}(r)\right) P_\ell(\cos\theta)[A_{\ell n}\cos(\omega_{\ell n} t) + B_{\ell n}\sin(\omega_{\ell n} t)]}$$

where $\zeta_n^{(\ell)}(r)$ includes the centrifugal term $\ell(\ell+1)/r^2$ absorbed into the effective potential.

7. d'Alembert / Flux Formulation

The probability flux density is $\Phi(r,t) = f(r,t)/(4\pi r^2)$. Imposing the d'Alembert condition on f:

$$\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial r^2} = 0$$

This is the **flat-space** (no R/r factor) 1D wave equation in r. Its general solution is:

$$f(r,t) = F(r - Vt) + G(r + Vt)$$

Applying $f(2R,0) = 0$:

$$F(2R) + G(2R) = 0 \Longrightarrow G(2R) = -F(2R)$$

A physically natural choice (outgoing wave only):

$$f(r,t) = F_0\left(e^{-\alpha(r-Vt)^2} - e^{-\alpha(r-2Vt)^2}\right), \quad \alpha > 0$$

or, for the standing-wave variant:

$$f(r,t) = A\sin\left(\frac{n\pi(r-2R)}{2R}\right)\cos(\omega_n t), \quad \omega_n = \frac{n\pi V}{2R}$$

For $f(R,0) = 0$, replace $2R \to R$:

$$f(r,t) = A\sin\left(\frac{n\pi(r-R)}{R}\right)\cos(\omega_n t), \quad \omega_n = \frac{n\pi V}{R}$$

8. Generalized Parameter Table

Case	Boundary	V	ω_n (Airy)	ω_n (d'Alembert)
I	$f(2R,0) = 0$	$c = 3 \times 10^8$ m/s	$k_n c$	$n\pi c/(2R)$
II	$f(2R,0) = 0$	$10^9 c$	$k_n \cdot 10^9 c$	$n\pi \cdot 10^9 c/(2R)$
III	$f(2R,0) = 0$	346 m/s	$k_n \cdot 346$	$n\pi \cdot 346/(2R)$
IV	$f(R,0) = 0$	c	$k_n^{(R)} c$	$n\pi c/R$

Case	Boundary	V	ω_n (Airy)	ω_n (d'Alembert)
V	$f(R,0)=0$	$10^9 c$	$k_n^{(R)} \cdot 10^9 c$	$n\pi \cdot 10^9 c/R$
VI	$f(R,0)=0$	346 m/s	$k_n^{(R)} \cdot 346$	$n\pi \cdot 346/R$

Physical interpretations: - $V = c$: photon / graviton propagation - $V = 10^9 c$: onduscular superluminal mode (TauTimp sector) - $V = 346$ m/s: acoustic / phonon mode

9. Numerical Parameters

$$R = 4.2355 \times 10^{22} \text{ m}$$
$$2R = 8.4710 \times 10^{22} \text{ m}$$

Airy zeros: $a_1 = 2.3381, \quad a_2 = 4.0879, \quad a_3 = 5.5206$

Ground-state wave number (Case A):

$$k_1 = \frac{a_1^{3/2}}{3\sqrt{2}\,R} = \frac{(2.3381)^{3/2}}{3\sqrt{2} \times 4.2355 \times 10^{22}}$$
$$k_1 \approx \frac{3.5764}{1.7970 \times 10^{23}} \approx 1.990 \times 10^{-23} \text{ m}^{-1}$$

Ground-state frequencies:

V	$\omega_1 = k_1 V$
$c = 3 \times 10^8$ m/s	5.97×10^{-15} rad/s
$10^9 c$	5.97×10^{-6} rad/s
346 m/s	6.89×10^{-21} rad/s

10. Flux Observable

The observable flux density:

$$\Phi(r,t) = \frac{f(r,t)}{4\pi r^2}$$

For the n-th Airy mode (Case A):

$$\Phi_n(r,t) = \frac{1}{4\pi r^{5/2}} \operatorname{Ai}(-\zeta_n(r))[A_n \cos(\omega_n t) + B_n \sin(\omega_n t)]$$

The **normalization condition** (probability interpretation):

$$4\pi \int_0^{2R} \Phi_n(r,t)\, r^2\, dr = \int_0^{2R} f_n(r,t)\, dr = 1$$

fixes the coefficients A_n in terms of the Airy function orthogonality integrals.

11. Asymptotic Behavior

For large ζ (WKB regime, $r \gg R^{1/3}/k^{2/3}$):

$$\operatorname{Ai}(-\zeta) \approx \frac{1}{\sqrt{\pi}\,\zeta^{1/4}} \sin\left(2/3\,\zeta^{3/2} + \pi/4\right)$$

So the field behaves asymptotically as:

$$f_n(r,t) \approx \frac{C_n}{\sqrt{\pi}\,r^{1/2}\,\zeta^{1/4}} \sin\left(2/3\,\zeta^{3/2} + \pi/4\right)\cos(\omega_n t)$$

which is a damped oscillation in r with effective local wave number $k_{\text{eff}}(r) = \sqrt{k_n^2 r/R}$.

End of derivation. See companion Maple transcript TauTimp_SphericalSymmetry.mpl for numerical computation and plots. Plot